

SHYAM A

Software Engineer | Machine Learning Engineer

[Contact](#) | [Mail](#) | [LinkedIn](#) | [GitHub](#) | [Portfolio](#) | Chennai, India

PROFESSIONAL SUMMARY

Software Engineer with internship experience designing, developing, testing, and deploying scalable web applications. Proficient in Java, Python, JavaScript, React.js, MySQL, Spring Boot, RESTful APIs, and Git, with a strong foundation in OOP, DSA, SDLC, and Agile development. Passionate about building reliable software solutions and continuously learning modern technologies.

WORK EXPERIENCE

Machine Learning Engineer Intern — [Grand Twin](#)

Feb 2026 – May 2026

- Designed and maintained end-to-end data preprocessing pipelines using Pandas and NumPy, reducing data noise by ~20% through outlier detection; conducted EDA with Matplotlib to uncover patterns and drive feature selection decisions.
- Engineered multi-stage feature engineering workflows covering encoding, scaling, and feature elimination; trained and evaluated Scikit-learn models, boosting predictive accuracy by ~15% and improving overall pipeline reliability.

Software Development Intern — [Chennai Metro Rail Limited \(CMRL\)](#)

Jan 2026 – Feb 2026

- Architected a secure RBAC dashboard with session-based auth across 3 user tiers and parameterized SQL queries; reduced manual retrieval time by ~30%, unauthorized access by ~40%, data entry errors by ~25%, and improved code maintainability by ~20% through modular backend design and active code reviews.

Junior Full-Stack Developer Intern — [Eagle Hi-Tech Soft Cloud Pt. Ltd.](#)

May 2025 – Jun 2025

- Developed, tested, and maintained production-ready full-stack features in Java and SQL within an Agile/Scrum SDLC; leveraged Git for version control, boosting team workflow efficiency by ~20% and reducing development rework by ~15%.
- Assisted in designing and implementing new application features, performed debugging and troubleshooting to resolve performance issues, and supported deployment pipelines aligned to project requirements and compliance standards.

PROJECTS

NIEVROX — SaaS Automation Platform | [Personal Project](#)

In Development

- Architecting a modular SaaS automation engine with lazy-loaded routes achieving ~22% bundle size reduction; building a responsive React/Next.js UI with Framer Motion and TypeScript, CI/CD pipelines, and scalable Node.js backend following.

Tech Stack: React, Next.js, TypeScript, Tailwind CSS, Framer Motion, Node.js, Git

Integrated Role-Based Dashboard System | CMRL

- Built a secure Flask + SQL dashboard with RBAC, JWT auth, and parameterized SQL queries; improved access efficiency by ~35%, cut unauthorized access by ~25%, and reduced backend latency by ~18% through query optimization and modular REST API design.

Tech Stack: Python, Flask, Flask-CORS, SQL, HTML5, CSS3

PROB-PSENN | Grand Twin [DEMO LINK](#)

- Developed an explainable AI model for MNIST digit classification using prototype-based learning, uncertainty estimation, and visual explanation techniques with a Gradio based Interactive Demo

Tech Stack: Python, PyTorch, Torch vision, CNN, Explainable AI(XAI), MC Dropout, Gradio, NumPy, Matplotlib, Seaborn, SciPy, Scikit-Learn, Pillow

TECHNICAL SKILLS

Languages: Java | Python | JavaScript | TypeScript | SQL | C | C++

Frontend: React.js | Next.js | HTML5 | CSS3 | Tailwind CSS

Backend: Node.js | Express.js | Flask | RESTful APIs | Spring Boot | JWT Authentication | SQL | NoSQL

Developer Tools: Git | GitHub | Version Control | Postman | VS Code | NPM

Cloud: Cloud Computing (NPTEL) | AWS Basics

Core: OOP | DSA | DBMS | Operating Systems | Computer Networks | Agile | SDLC | Unit Testing

EDUCATION

Bachelor of Computer Applications (BCA)

2023 – 2026

St. Thomas College of Arts and Science, Chennai | CGPA: 8.0 / 10.0

CERTIFICATIONS

- NPTEL — Cloud Computing (IIT)
- Infosys — Python Basics & OOP with Python
- Microsoft — Artificial Intelligence & Data Science
- Excel IR EdTech — Salesforce Development